Frontier Post-Training Recipe Survey

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From the classic 3-step RLHF recipe to the rise of multi-teacher on-policy distillation (MOPD).

Canonical recipes over time in post-training

The shape of a post-training recipe has changed more in the last year than in the prior three.

- **2022–2023 (InstructGPT):** one pipeline — SFT → reward model → RL.
- **2024 (Llama 3, Tülu 3, etc.):** open recipes formalize SFT → DPO → RL with verifiable rewards. Closed recipes use many stages of RLHF.
- **2025 (DeepSeek R1):** reasoning RL (R1) makes large-scale RL the centerpiece.
- **2026 (MiMo Flash V2):** recipes fragment into *many specialist models* that are merged back into one.

The new thing: MOPD

Multi-teacher On-Policy Distillation (MOPD) is the pattern showing up across the 2026 frontier.

1. Train **N domain-specialist teachers** (each: SFT, then RL on the relevant domains).
2. Train **one general student** by sampling *its own* trajectories (this is the final post-trained model).
3. On each rollout, minimize **reverse-KL** to the *relevant* teacher's output distribution, token by token.

Lineage: MiMo Flash v2 introduced it → DeepSeek V4 & Nemotron 3 Ultra scale it to >10 teachers.

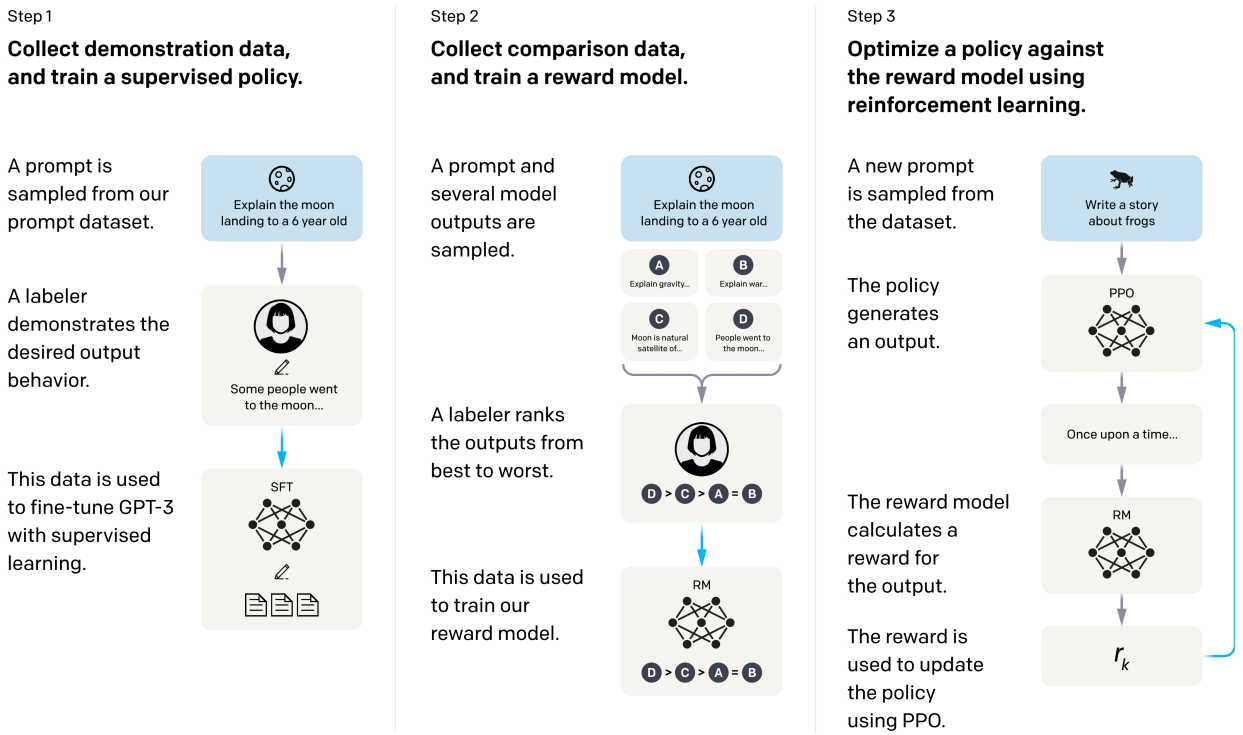
Why did MOPD emerge?

- **RL got expensive and conflict-prone.** Mixing math, code, and agentic RL in one run eventually trades capabilities off against each other.
- **Specialists are cheap to make / organizationally scalable.** SFT-then-RL on a single domain is well understood and parallelizable. As post-training becomes more complex, scaling it across organizations is a big win.
- **On-policy distillation matured.** Literature and know-how continued to emerge through the RLVR renaissance.

Source: DeepSeek V4 §5.1, MiMo-V2-Flash

The path to today

InstructGPT (Mar. 2022) — the canonical 3 steps

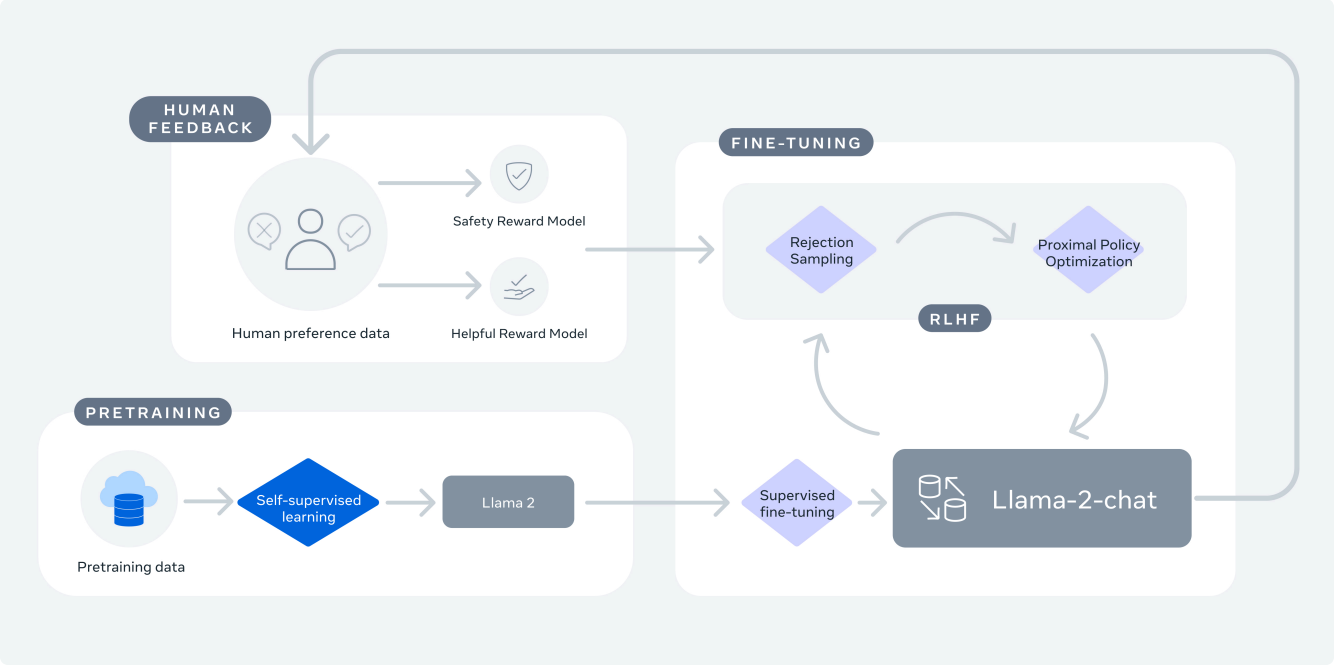


- **SFT** on human demonstrations
- **Reward model** trained on human comparisons
- **PPO** against the reward model

InstructGPT: SFT on demonstrations → reward model on comparisons → PPO

Source: InstructGPT (arXiv:2203.02155)

Llama 2 (Jul. 2023) — multi-stage RLHF

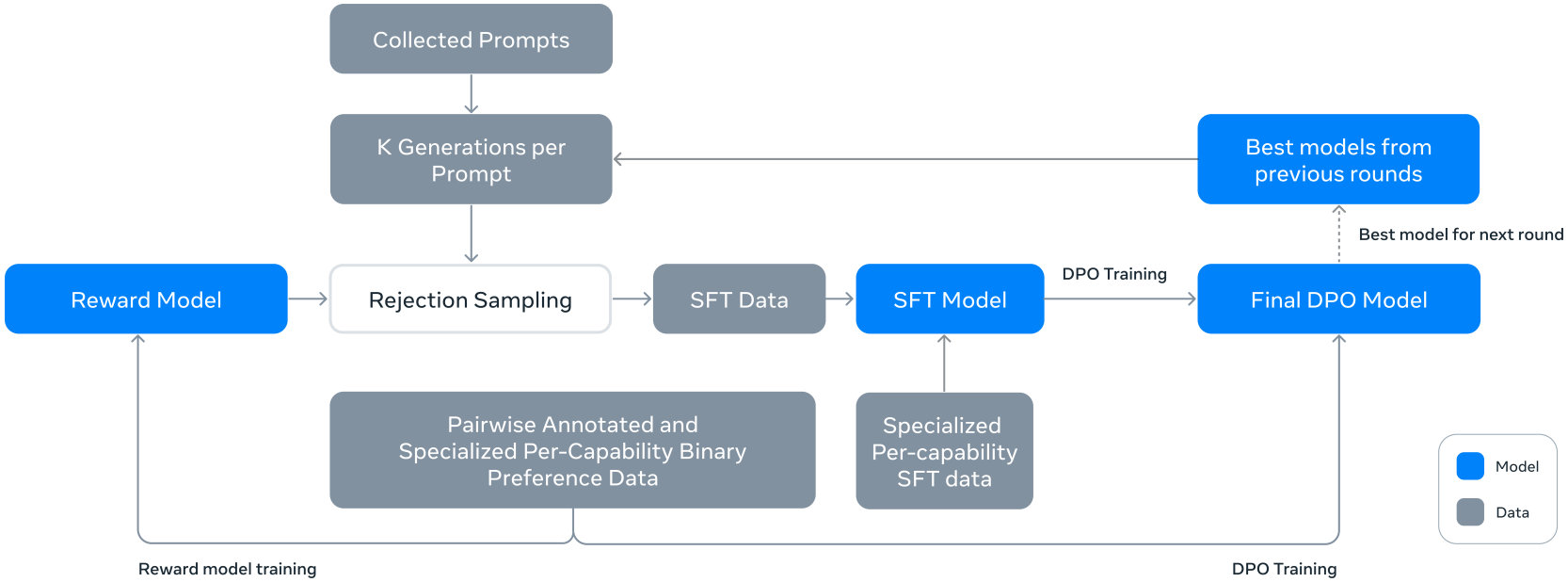


Llama 2: pretrain → SFT → iterative RLHF with rejection sampling and PPO

- **SFT**, then **iterative RLHF** over multiple rounds
- Each round: **rejection sampling → PPO**
- **Two** reward models — separate **helpfulness** and **safety**

Source: Llama 2 (arXiv:2307.09288)

Llama 3 (Jul. 2024) — a complex multi-stage recipe with simpler optimizers

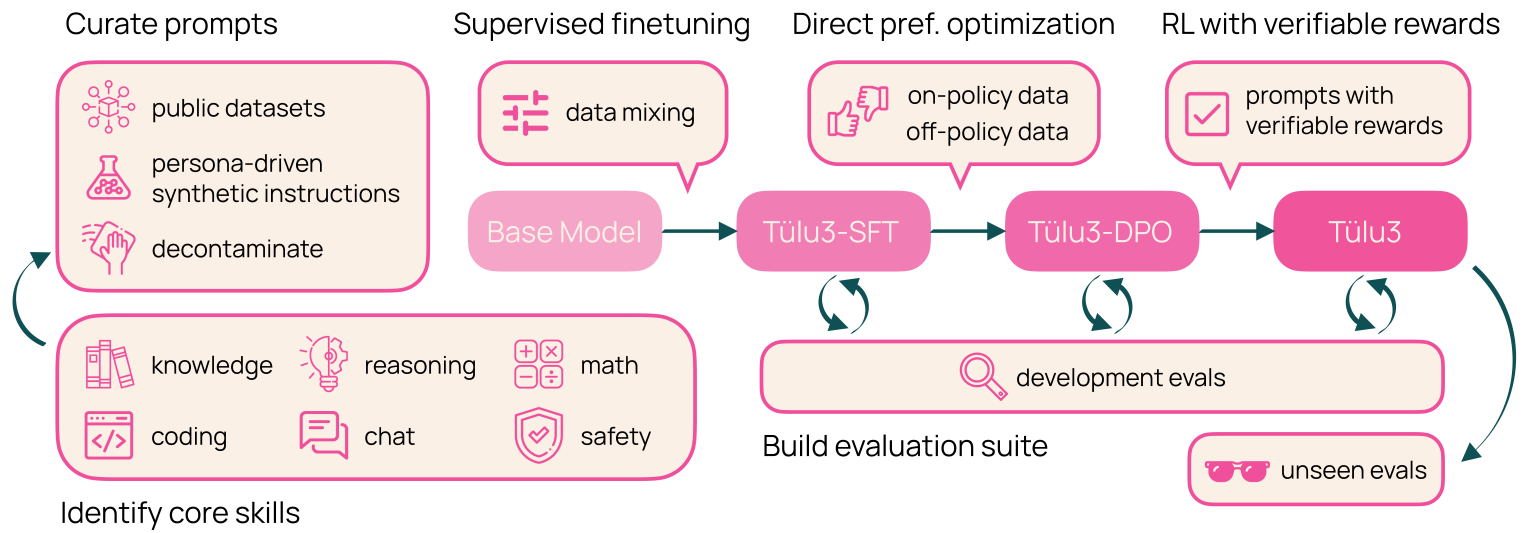


Llama 3 post-training: reward model → rejection sampling → SFT → DPO, iterated over rounds with best models feeding the next

- Per round: **reward model** → sample **K per prompt** → **rejection sampling** → **SFT** → **DPO**
- No online RL — the RM only **filters**; run over **6 rounds**, best models seed the next

Source: Llama 3 (arXiv:2407.21783)

Tülu 3 (Nov. 2024) — simple three-stage post-training



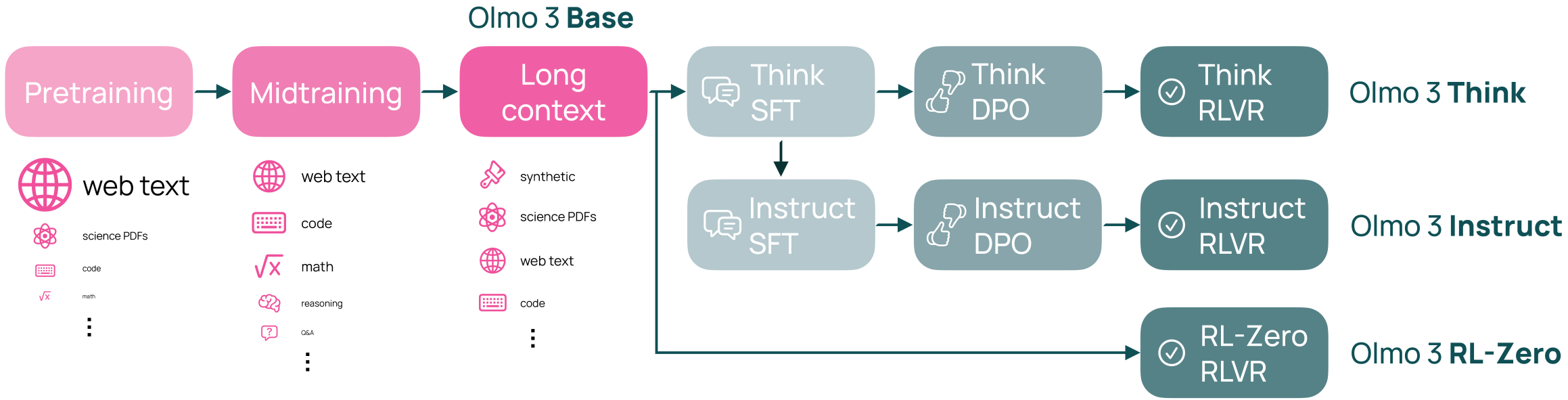
Tülu 3: curate prompts → SFT → DPO → RLVR with a held-out eval suite

Curated prompts → **SFT** → **DPO** → **RLVR**

(RL with verifiable rewards – the acronym was coined in this paper)

Source: Tülu 3 (arXiv:2411.15124)

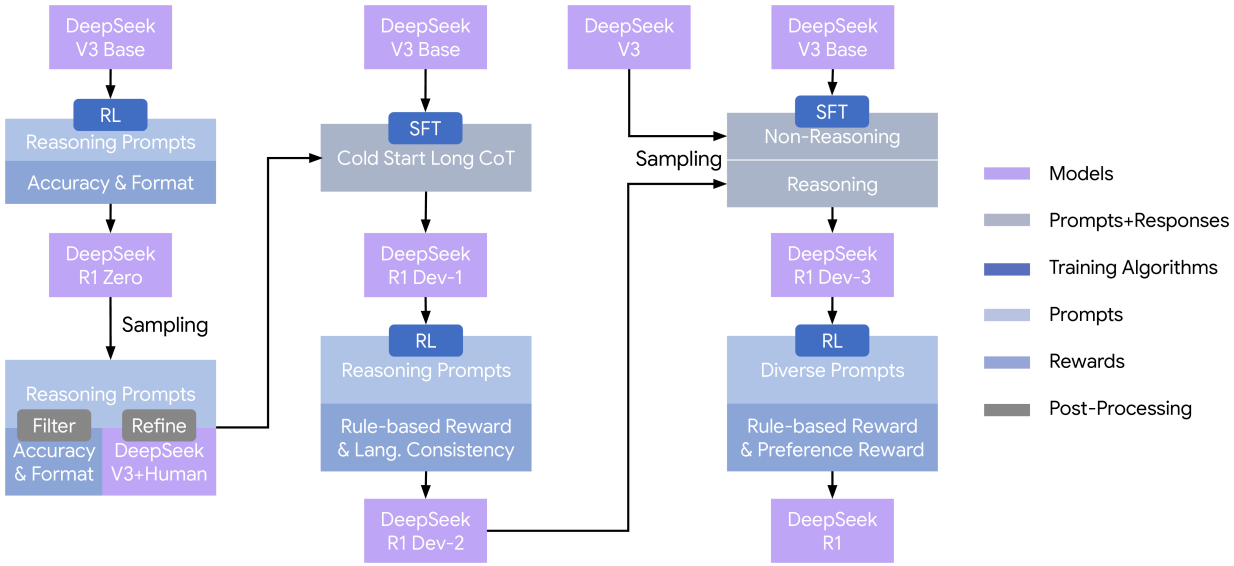
Olmo 3 (Dec. 2025) — a reasoning update to the Tülu 3 recipe



Olmo 3 model flow: Pretraining → Midtraining → Long context, then Think / Instruct / RL-Zero branches each SFT → DPO → RLVR

Source: Olmo 3 (arXiv:2512.13961)

DeepSeek R1 (Jan. 2025) — RL as the centerpiece



DeepSeek-R1 multi-stage pipeline: R1-Zero, then cold-start SFT → reasoning RL → rejection-sampling SFT → final RL

The recipe:

- **R1-Zero** — pure RL (GRPO) on the base, *no SFT*; used to **seed reasoning behaviors** for the full run, not a separate product
- **R1** — cold-start SFT → reasoning RL → rejection-sampling SFT → final RL → distill to dense
- **A big change in recipes:** Large-scale RLVR as the primary driver, SFT to distill and refine RL behaviors

Part of the DeepSeek line. Source: DeepSeek R1 (arXiv:2501.12948) · Interconnects: R1 recipe for o1

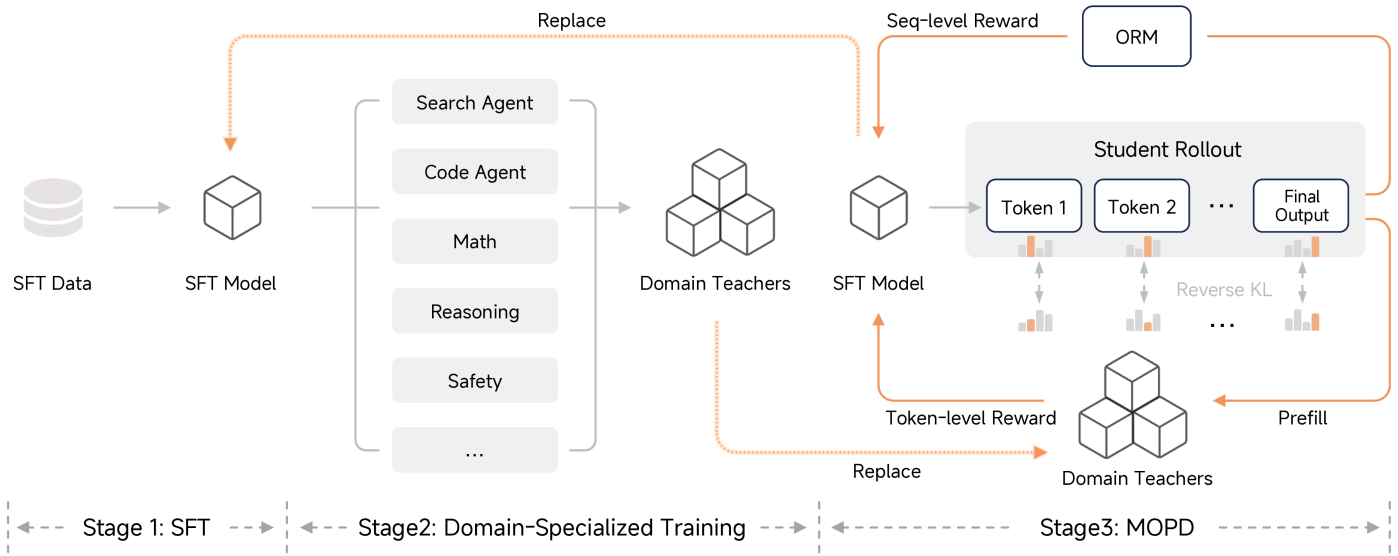
DeepSeek evolution after V3

- **V3** · Dec '24 — SFT + GRPO RL.
- **R1** · Jan '25 — multi-stage RL; reasoning *emerges*.
- **V3.1** · Aug '25 — hybrid think / non-think in one model.
- **V3.2** · Dec '25 — 6 specialists via RL → SFT distillation → one mixed GRPO.
- **V4** · Apr '26 — 10+ domain experts → MOPD.

Each model links to its report. V3.1 detail is from its model card (no full report); V3-0324 was a quality bump on the same recipe, so it's omitted.

2026-style recipes!

MiMo Flash v2 (Jan. 2026) — where MOPD started



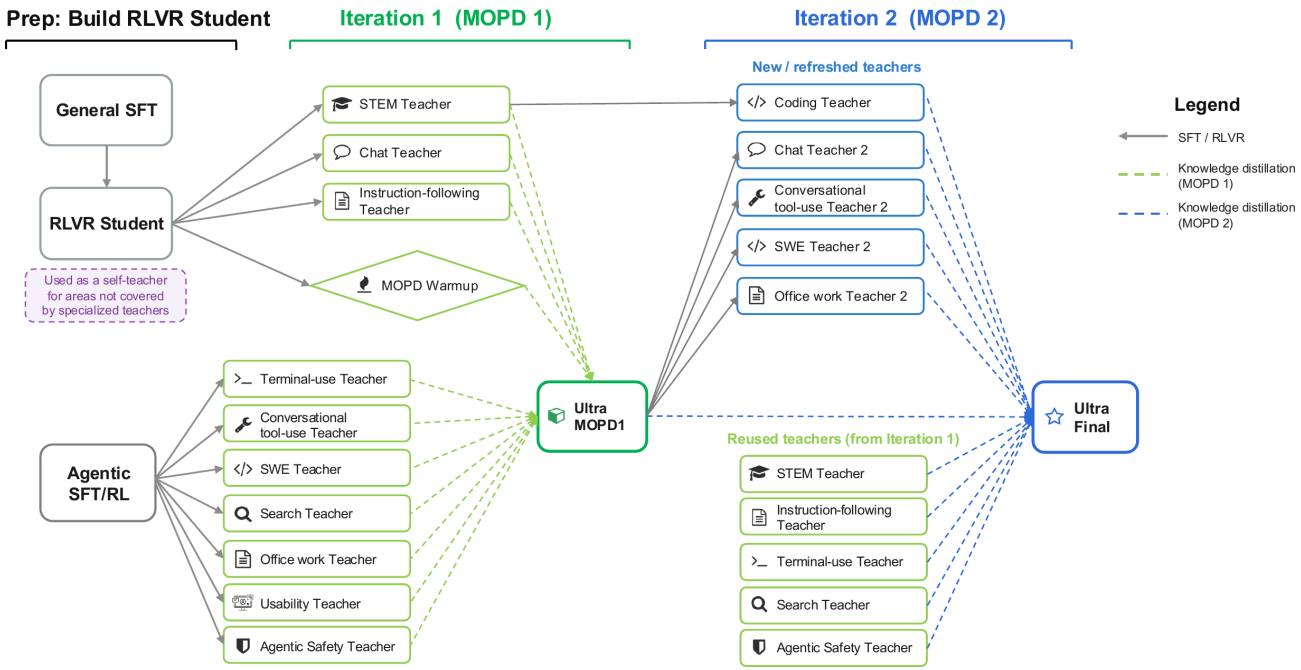
MiMo Flash v2 post-training: SFT → domain teachers → multi-teacher on-policy distillation

Stages: Stage 1 SFT → Stage 2 train ~6 domain-specialist teachers (with older style post-training recipes) → Stage 3 MOPD into a single student.

First clean articulation of **multi-teacher on-policy distillation** as the consolidation step — replaces a single monolithic RL stage with distill-from-specialists.

Source: MiMo-V2-Flash report (arXiv:2601.02780)

Nemotron 3 Ultra (Jun. 2026) — two rounds, many teachers



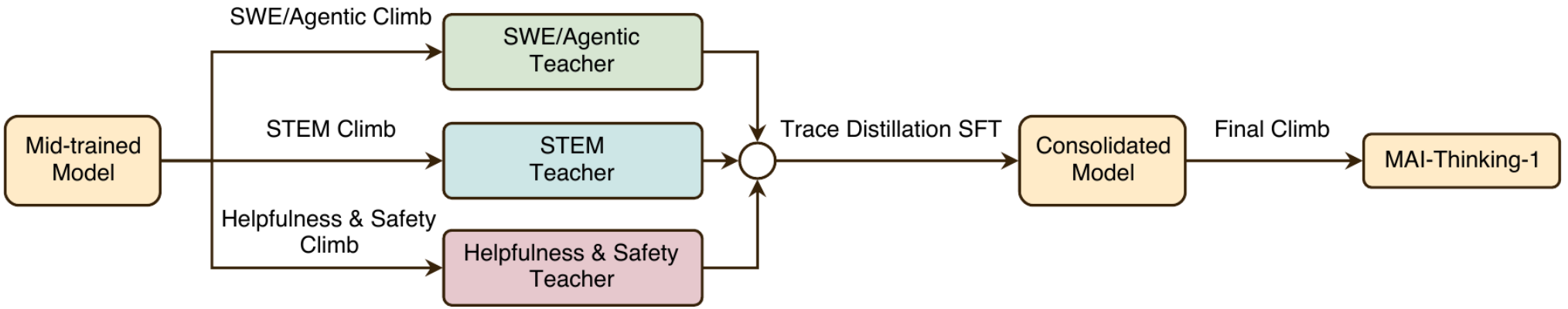
Nemotron 3 Ultra: two-iteration multi-teacher on-policy distillation

Stages: SFT → **multi-teacher on-policy distillation**, run over **two iterations**, with **>10 teachers** spanning reasoning, code, math, and agentic domains.

Novel: multi-round MOPD across different domains — distill, then re-distill from refreshed teachers.

Source: NVIDIA Nemotron 3 Ultra technical report

MAI-Thinking-1 (Jun. 2026) — closer to R1 than V4



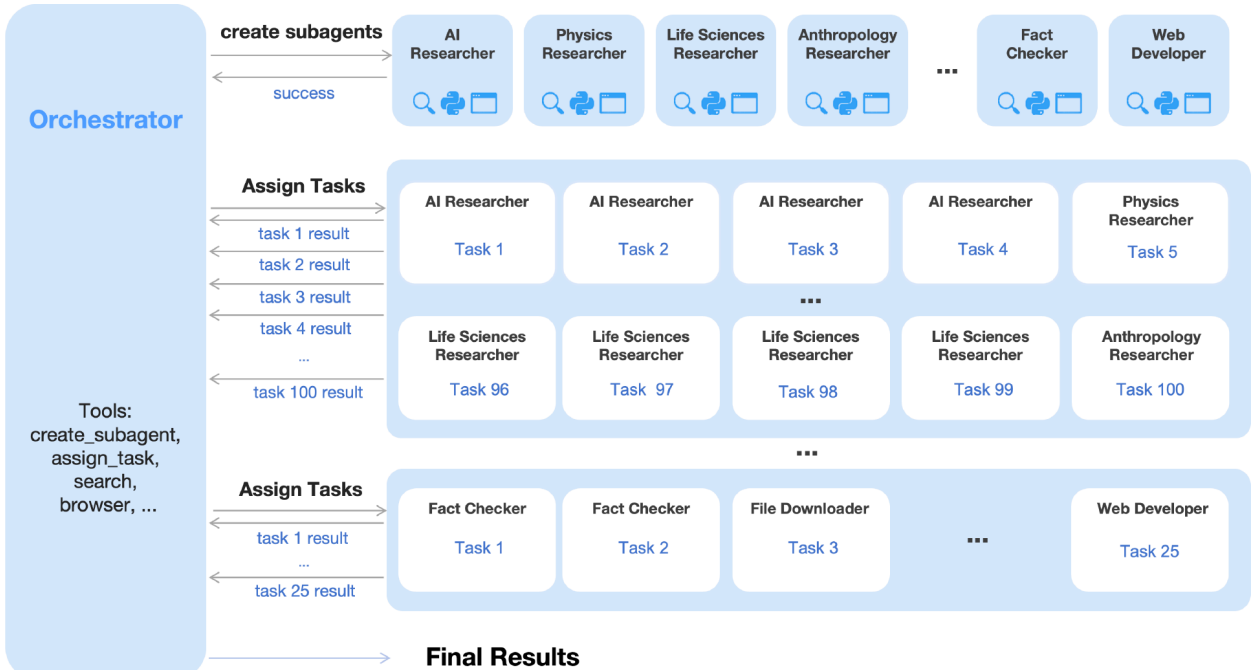
MAI-Thinking-1: specialist RL climbs → trace-distillation SFT → consolidate → final climb

Stages: mid-trained base → **3 specialist RL "climbs"** (e.g. STEM) → **trace-distillation SFT** to consolidate the climbs → a final RL climb → MAI-Thinking-1.

Closer to DeepSeek R1 than to V4 — multi-stage RL with trace-distillation SFT to consolidate, *not* on-policy MOPD. Not the only lab without MOPD!

Source: MAI-Thinking-1

Kimi K2.5 (Jan. 2026) — agentic, multimodal



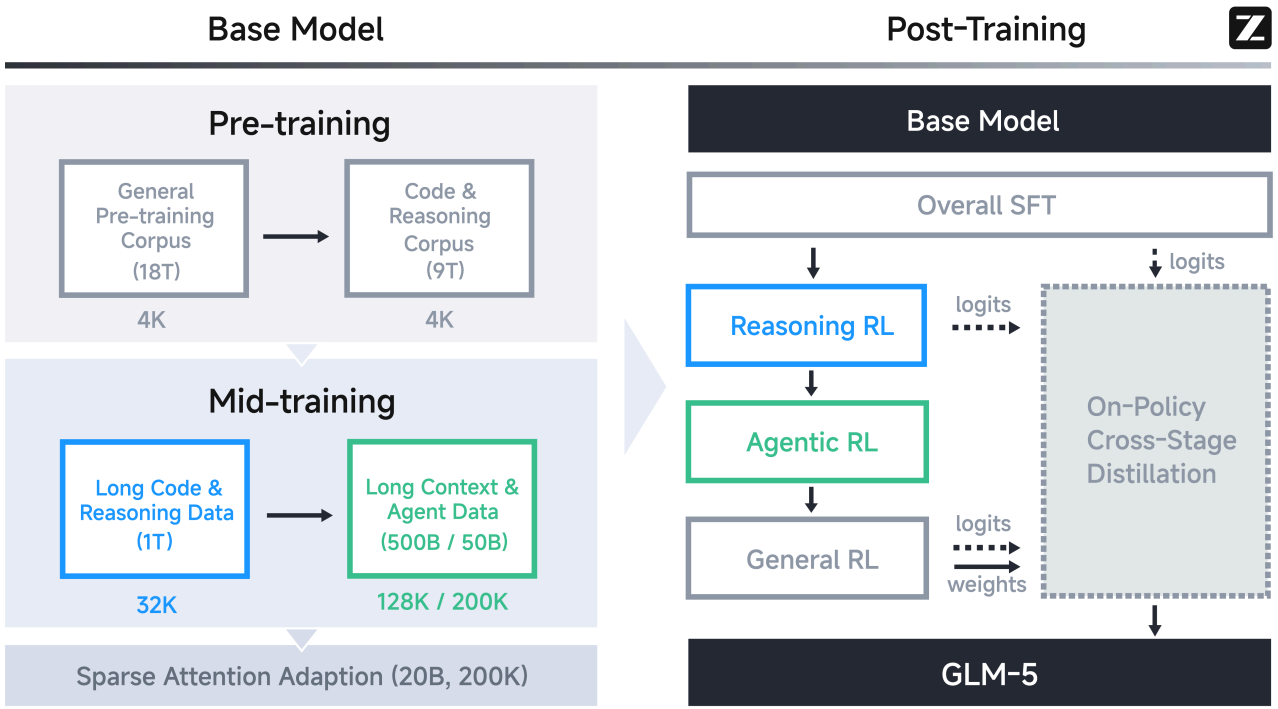
Kimi K2.5 Agent Swarm: self-directed parallel agent orchestration

Stages: text-only SFT → joint text–vision RL across coding, vision, reasoning, agentic tasks.

(No mention of MOPD)

Source: Kimi K2.5 technical report · blog

GLM-5 (Feb. 2026) — staged RL by capability



GLM-5 pipeline: Base → SFT → Reasoning RL → Agentic RL → General RL with cross-stage distillation

Stages: Base → SFT → **Reasoning RL** → **Agentic RL** → **General RL**.

MOPD isn't universal yet but it's surging!

Source: GLM-5 technical report (arXiv:2602.15763)

So, where are we going from here?